

Reader Report: Week 06

Probabilistic Invariance: VI-BIP and the Core Theorems

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Summary

Wu and Blei (2025) place a prior over binary selectors $z \in \{0, 1\}^p$, where $z_j = 1$ indexes a feature included in the invariant set S^* (Peters et al., 2016). The posterior $p(z \mid \mathcal{D}) \propto p(z) \prod_{e,i} \hat{g}(Y_i^e \mid X_{ei}^z) / \hat{p}_e(Y_i^e \mid X_{ei}^z)$ rewards selectors for which the pooled conditional $\hat{g}$ dominates each local conditional $\hat{p}_e$, where $\hat{p}_e$ is fit separately per environment e . Consistency (Thm 1) requires $\mu_{\min} > 0$, defined as the minimum KL divergence between $\hat{g}$ and any $\hat{p}_e$ over non-invariant features (Wu and Blei, 2025, Def. 1). Contraction (Thm 2) gives posterior mass on $\{z \neq z^*\}$ scaling as $O(R e^{-\kappa n E \mu_{\min}})$, where $R = 2^p$ counts hypotheses, E is the number of environments, and κ is a universal constant. VI-BIP (Wu and Blei, 2025, Sec. 4) approximates the posterior via mean-field variational inference (Blei et al., 2017), optimizing the evidence lower bound (ELBO) using the U2G gradient estimator to handle discrete z .

Critique: One Invariance Condition, Three Algorithms

The most striking observation across this body of work is that Peters et al. (2016), Fan et al. (2024), and Wu and Blei (2025) all enforce the invariance condition $P_e(Y \mid X_{S^*}) = \text{const}$ for all $e \in \mathcal{E}$, yet operationalize it through algorithms that produce different outputs: a confidence set, a point estimate, and a posterior distribution, respectively.

Peters et al. test invariance sequentially, building a confidence set $\hat{S}(\mathcal{E})$ that shrinks monotonically as environments accumulate (Peters et al., 2016, Thm. 1). BIP’s Thm 2 is the probabilistic counterpart: rearranging the contraction gives sample complexity $nE \gtrsim O(p/\mu_{\min})$, which is the per-environment requirement for the Peters test to have nontrivial power. BIP thus does not replace ICP; it quantifies ICP’s frequentist power curve in a Bayesian language.

NegDRO (Fan et al., 2024) makes a third move. Its minimax problem $\min_b \max_{w \in U(\gamma)} \sum_e w_e \mathbb{E}[(Y^e - b^\top X^e)^2]$, where $U(\gamma)$ is the set of signed environment weights with total variation at most γ (Fan et al., 2024, Def. 1), is the distributionally robust framing of Bühlmann (2020) pushed to an adversarial extreme. BIP treats environments symmetrically in the product likelihood; NegDRO can amplify a single harsh environment pair via negative weights, making it more powerful when one intervention is known *a priori* to be dominant. The two methods have different assumptions about what the analyst knows before seeing the data, which is a structural incompatibility neither paper acknowledges.

A further issue concerns benchmarking. Wu and Blei (2025, Sec. 5) validate VI-BIP on a single genomics dataset: 6,170 yeast genes across CRISPR knockout conditions where $\mu_{\min} > 0$ holds by construction. The paper provides no evaluation in low-environment or observational settings, where $\mu_{\min}$ is unknown; whether the contraction advantage over ICP survives soft-intervention regimes is unresolved.

Questions

1. NegDRO’s λ and BIP’s $\mu_{\min}$ both quantify environment heterogeneity but measure different objects: λ is a spectral condition on second moments of X , while $\mu_{\min}$ is a KL condition on $Y \mid X$ conditionals. Is it possible to certify $\mu_{\min} > 0$ from observational data without already knowing z^* , or does this require the hard interventions that make the genomics setting tractable?
2. Wu et al. propose a size-restricted prior with parameter $p_{\max}$ to limit BIP’s support to subsets of size at most $p_{\max}$. Could NegDRO’s point estimate $\hat{\beta}^\gamma$ serve as the pre-screening filter, setting $z_j = 0$ for features with $|\hat{\beta}_j^\gamma| < \varepsilon$ to reduce the effective model count from 2^p to $2^{p'}$, and does this avoid the double-dipping problem that arises from using the same data to form both the prior and the likelihood?

References

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